Bellabeat Case Study

Activity & Sleep Patterns Among Fitbit Users

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Executive Summary

Bellabeat is a wellness technology company that designs health-tracking devices for women. This analysis examines smart device usage data from Fitbit users to identify behavioral trends that Bellabeat can apply to its own product and marketing strategy. Using activity and sleep records from 33 users over a one-month period, the analysis found that sleep is inconsistently tracked (logged on only 43.8% of days), daily activity is logged far more consistently (in 91.8% of days), average step counts fall short of the CDC's 10,000-step benchmark, and total steps have a statistically significant, if weak, negative relationship with sleep duration — while calories burned show no meaningful relationship with sleep at all. These findings point toward specific, actionable opportunities for Bellabeat to increase engagement with sleep tracking and to message around activity timing rather than calorie burn.

Business Task

Bellabeat's marketing team wants to better understand how people already use non-Bellabeat smart devices, to identify opportunities for Bellabeat's own product line and marketing strategy. This analysis uses publicly available Fitbit fitness tracker data as a proxy for broader smart-device usage trends, focusing on the relationship between daily activity, sleep tracking, and sleep duration.

Data Source & Preparation

Data: Fitbit Fitness Tracker Data (Kaggle, via Mobius) daily activity and sleep records from 33 Fitbit users, collected via Amazon Mechanical Turk. Case study brief: Google Data Analytics Professional Certificate (Coursera).

The daily activity table and the sleep table were merged on user Id and ActivityDate to create a single combined table for analysis in Google BigQuery.

Data cleaning: duplicate row issue

During validation, cross-checking correlation results between BigQuery and Tableau surfaced a discrepancy: BigQuery returned 413 rows where Tableau returned 410. Investigating the mismatch revealed 3 duplicate Id + ActivityDate combinations in the merged table — the same user-day appearing twice with identical values, likely introduced during the join. These duplicates were inflating counts and skewing computed correlations. All analysis below uses a deduplicated dataset (SELECT DISTINCT on Id, ActivityDate, and the relevant metric columns), reducing the working dataset from 413 to 410 unique daily records. This step was essential: the uncorrected data produced a misleadingly significant calories-sleep relationship ($p = 0.0105$) that disappeared entirely once duplicates were removed ($p = 0.522$), while the true steps-sleep relationship — initially masked — emerged as statistically significant only after cleaning.

Methodology

Analysis was performed in Google BigQuery using SQL (aggregation, CASE statements, and the CORR() function for Pearson correlation coefficients). Visualizations and trend lines were built in Tableau Public, cross-referenced against the BigQuery results to confirm consistency. Full queries are included in the accompanying bellabeat_queries.sql file. Analysis was supported by Claude (Anthropic), used as an AI assistant for query debugging guidance, statistical explanation, and report drafting support.

Key Findings

1. Sleep is tracked on fewer than half of activity days

Of 943 total activity-logged days, sleep data was recorded on only 413 days (43.8%) and missing on 530 days (56.2%). This is the single biggest data-quality gap in the dataset and directly limits how much Bellabeat (or any wellness platform) can act on sleep-related insights for a given user.

2. Most days include logged activity

866 of 943 days (91.8%) show logged activity, versus 77 days (8.2%) with zero recorded activity. Users are engaging consistently with activity tracking — the gap is specifically in sleep tracking, not tracking in general.

3. Average daily steps fall short of the CDC benchmark

The average across users was 8,333 steps per day, below the CDC's commonly cited 10,000-step benchmark for general fitness. This is a meaningful, actionable gap rather than a marginal shortfall.

4. Calories show no meaningful relationship with sleep duration

Correlation between daily calories burned and total minutes asleep was effectively zero ($r = -0.0317$, $R^2 = 0.001$, $p = 0.522$, $n = 410$). The relationship is not statistically significant — calories burned in a day do not predict how much someone sleeps that night in this dataset.

5. Steps show a weak but statistically significant relationship with sleep

Total daily steps had a weak negative correlation with sleep duration ($r = -0.1903$, $R^2 = 0.036$, $p = 0.0001$, $n = 410$). Statistically significant, meaning the relationship is unlikely to be due to chance — though the effect size remains modest, so steps alone explain only a small share of the variation in sleep.

Finding	Metric	Result	Significant?
Sleep tracking rate	% days logged	43.8% (413/943)	—
Activity tracking rate	% days logged	91.8% (866/943)	—
Avg. steps vs. benchmark	Steps/day	8,333 vs. 10,000 target	—
Calories × Sleep	Pearson r	-0.0317	No (p = 0.522)
Steps × Sleep	Pearson r	-0.1903	Yes (p = 0.0001)

Recommendations

1. Prioritize closing the sleep-tracking gap. With sleep logged on under half of days, Bellabeat should invest in reminders, simplified logging, or incentives (streaks, badges) specifically for sleep — without consistent sleep data, neither users nor Bellabeat can act on sleep-related insights.
2. Message around activity timing and consistency, not calorie burn, when discussing sleep. Since calories show no relationship with sleep but steps do, marketing and in-app guidance aimed at improving sleep should focus on daily movement and timing, not calorie targets.
3. Introduce personalized step goals to close the benchmark gap. With average steps at 8,333 versus the 10,000 target, gamification, milestone nudges, or personalized goal-setting could help close a meaningful and measurable activity gap.
4. Treat the steps-sleep relationship as a starting point, not a full explanation. The relationship is real but weak ($R^2 = 0.036$) — Bellabeat should continue to investigate other factors (e.g., stress, screen time, caffeine) that may explain more of the variation in sleep duration.

Limitations

- Small sample: 33 users, which limits how confidently these findings generalize to a broader population.
- Short observation window: approximately one month of data (April–May 2016).
- Self-reported/wearable data via Fitbit, not Bellabeat's own devices — usage patterns may differ on Bellabeat hardware.

- Correlation does not imply causation; the steps-sleep relationship is statistically significant but does not establish that increasing steps causes better or worse sleep.
- A data-quality issue (3 duplicate rows from the table join) was identified and corrected during this analysis; all reported results reflect the deduplicated 410-record dataset.
- No demographic data (including gender) is available in this dataset. Since Bellabeat's products are designed specifically for women, and this dataset does not confirm the gender of any user, findings should be interpreted as general smart-device usage patterns rather than patterns specific to Bellabeat's target audience.

Supporting Materials

Interactive dashboard and story: [https://public.tableau.com/views/Bellabeat-ActivitySleepPatterns/BellabeatCaseStudy?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link]

SQL queries: bellabeat_queries.sql (included alongside this report)